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The advanced copper oxide coating separator enhances the electrochemical performance of high-rate lithium-ion batteries

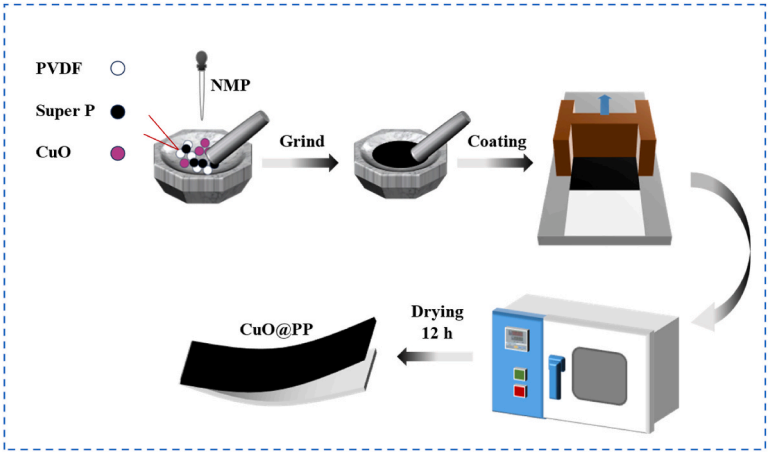
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HIGHLIGHTS

- High-performance CuO@PP separator is fabricated using a simple method.
- The battery performances with the CuO@PP separator are remarkably enhanced.
- The CuO@PP separator boosts cathode capacity by about 10 mAh g^{-1}.

GRAPHICAL ABSTRACT



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ABSTRACT

The performance and cycle life of Lithium-ion batteries (LIBs) are significantly influenced by the separator, a critical component. Traditional polyolefin separators can no longer meet the growing demand for energy storage. Coating conventional polyolefin separators with electrochemically inert ceramic materials can enhance stability but often increases weight and reduces battery capacity. This study introduces a novel separator coated with copper oxide (CuO), developed through a simple and efficient method. The CuO coating polypropylene (CuO@PP) separator demonstrates superior thermal resistance, enhanced electrolyte absorption, and increased porosity compared to conventional polypropylene (PP) separators. Additionally, the active participation of the

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CuO coating in electrochemical reactions during the charge-discharge process increases battery capacity. Results indicate that the CuO@PP separator provides an additional capacity of approximately 10 mAh g^{-1}. The battery using the CuO@PP separator exhibited exceptional electrochemical properties in contrast to the traditional PP separator, with a first discharge capacity of 129.8 mAh g^{-1} at 10 C and a capacity retention rate of 92.4 % over 400 cycles at 10 C. Thus, enhancing the reliability and performance of LIBs by modifying the PP separator with a CuO coating presents a promising approach, offering novel perspectives for designing and manufacturing advanced battery separators.

1. Introduction

Lithium-ion batteries (LIBs) are extensively utilized in contemporary technology and daily life. They serve as the primary power source for electric vehicles, smartphones, laptops, and are also integral to energy storage systems, renewable energy generation, and grid balancing. Their low self-discharge rate, high energy density, and long cycle life make them ideal for portable electronic devices [1–6]. LIBs comprise a cathode electrode, an anode electrode, an electrolyte, a separator, and a casing. Among these components, the separator is crucial for the cycle life and safety of LIBs. The primary function of the LIB separator is to isolate the anode and cathode electrodes, preventing short circuits. Simultaneously, it allows lithium ions to pass through, maintaining electrochemical balance within the battery, thereby ensuring both safety and performance [7–9]. Although the separator is not electrochemically active and does not participate in electrochemical reactions, it determines the interface between the electrolyte and the electrode, significantly impacting battery performance [10–12]. For example, improved wettability of the separator with the electrolyte enhances charge transfer across the interface, improving discharge capacity and rate capability [13,14]. Consequently, developing high-performance separators is a substantial challenge.

Polyolefin separators have become the predominant choice for separators in LIBs because of their low cost, high mechanical strength, and excellent chemical resilience [15–18]. Nevertheless, conventional commercial separators exhibit high interfacial impedance, poor thermal stability, and low wettability, making them susceptible to short circuits, which can cause serious safety problems [19]. Researchers have proposed various approaches to address these problems. These approaches include using ceramic materials [20,21], constructing metal-organic frameworks [22], and incorporating polymer blends [23] or multilayers [24] to modify the surface or structure of polyolefin separators. While these modifications may enhance the performance of polyolefin separators in terms of hydrophilicity, mechanical strength, and thermal stability, most of these modified materials are inactive and do not enhance battery capacity. They may even increase the thickness and weight of the separator, leading to a decrease in electrochemical performance [25]. Recently, Modeste et al. prepared a $LiFePO_4$-coated separator and assembled a $LiFePO_4$/Li battery using this separator. The interaction between the separator and $LiFePO_4$ is enhanced by the $LiFePO_4$ coating, which also contributes additional capacity [22]. Therefore, we can add active coatings to the separator to contribute additional capacity.

Hence, in this study, a CuO@PP separator was fabricated by coating CuO particles to enhance the electrochemical properties of LIBs (Fig. 1). CuO has emerged as a promising electrode material due to its excellent thermal stability, cost-effectiveness, environmental friendliness, and ability to enhance surface stability under high voltage conditions. Moreover, CuO is recognized for its high chemical stability and strong adhesion to various substrates, which contribute to maintaining the structural integrity of the coated materials. Its multifunctional properties, which enhance both safety and performance, make CuO a preferred choice for coatings in energy storage systems [26–30]. By coating a conventional PP separator with CuO particles, we designed an advanced separator that improves electrolyte wettability and increases capacity by engaging in the electrochemical reactions throughout the charge-discharge cycle. While most previous investigations have focused on the CuO coating of separators on the negative electrode, our research targets the CuO coating on the cathode side [31,32]. Experimental results indicate that the CuO-coated separator offers multiple benefits compared to the PP separator, including superior electrochemical performance, increased heat resistance, and improved electrolyte absorption. Furthermore, the CuO-coated separator exhibits superior discharge capacity relative to the PP separator when subjected to high discharge rates. This research offers significant perspectives into the development and production of advanced battery separators.

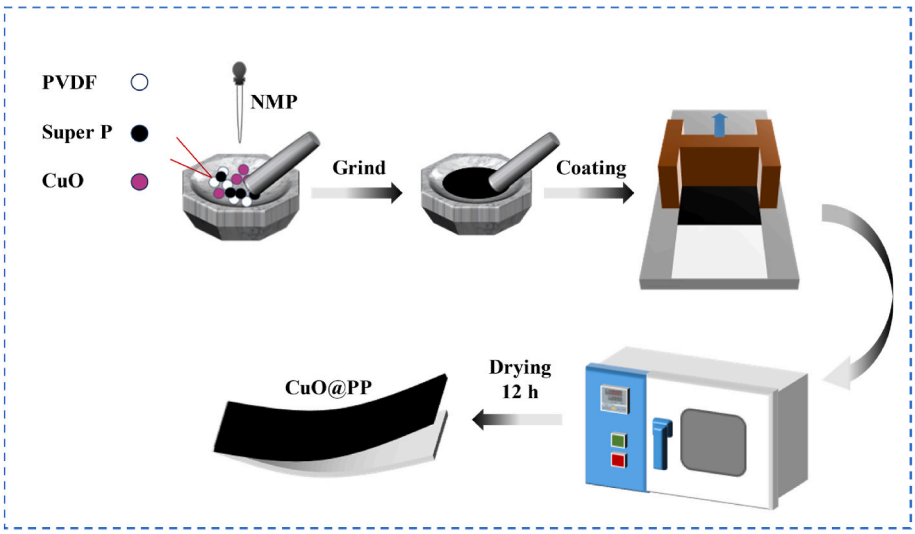


Fig. 1. Schematic fabrication process of the CuO@PP separator.

2. Experimental section

2.1. Preparation of CuO@PP separator

CuO powders are obtained from Aladdin Biochemical Technology Co., Ltd. These powders have a D_{50} particle size distribution of 10 μm. The microporous polypropylene (PP) separator (model Celgard 2400) is sourced from Celgard, LLC. The PP separator has a thickness of 25 μm. The slurry comprised 90 wt% Copper Oxide (CuO), 5 wt% Super P, and 5 wt% polyvinylidene fluoride (PVDF). PVDF was dissolved in N-methyl-2-pyrrolidone (NMP), followed by the addition of Super P and CuO powder. The mixture is ground for 30 min and subsequently coated onto a single surface of a PP separator with a doctor blade. The coated separator was then dried for 12 h in a vacuum oven set to 60 °C. The CuO-coated separators, with a coating thickness of 10 μm, were denoted as CuO@PP.

2.2. Characterization and measurements of CuO@PP separator

A scanning electron microscope (SEM, Rigaku SmartLab SE, Japan) operating at 10 kV was used to observe the morphologies of the CuO@PP separator. The mechanical behavior of the separator was measured by tensile tests at room temperature at a tensile rate of 10 mm/min on a CMT8102 tensile testing machine. The contact angle of the electrolyte on the separator surface was evaluated to examine the wettability of the separators. The separator, with a diameter of 19 mm, was treated in an oven at 60–150 °C for 30 min, and the area change before and after heating was measured to evaluate thermal shrinkage. To measure the porosity of the separators, the samples were soaked in n-butanol for 2 h and estimated using an equation (1) [33]:

$$Porosity(\%) = \frac{M_1 - M_0}{\rho \times V} \times 100\% \quad (1)$$

where M_0 and M_1 are the weight of the separator before and after soaking in n-butanol, respectively; ρ represents the density of n-butanol; V represents the volume of the separator. The electrolyte uptake of the separators was determined by measuring the weight difference before and after a 2-h immersion in the electrolyte, and calculated using equation (2) [14]:

$$Electrolyte\ uptake(\%) = \frac{M_1 - M_0}{M_0} \times 100\% \quad (2)$$

where M_0 and M_1 are the weight of the separator before and after soaking in electrolyte, respectively.

2.3. Electrochemical measurements

The cathode is prepared using a slurry coating method. $LiFePO_4$, PVDF, and Super P were dissolved in N-methyl-2-pyrrolidone (NMP) at a weight ratio of 80:10:10 to form a uniform slurry. This slurry iis then coated onto aluminum foil and dried at 80 °C under vacuum for 12 h. The liquid electrolyte consisted of 1 M $LiPF_6$ in a mixture of ethylene carbonate (EC), dimethyl carbonate (DMC), and ethyl methyl carbonate (EMC) in a 1:1:1 vol ratio. The battery is prepared in an argon-filled glove box with either a PP separator or a CuO@PP separator. Constant current charge and discharge tests are performed at 25 °C using a battery test system (BTS3000, Neware, China) at different current densities (1C = 170 mAh g^{-1}) in the voltage range of 2.0–4.2 V (vs. Li^+/Li). Cyclic voltammetry (CV) analyses are measured using an electrochemical workstation (Chi760E, China) within a potential range of 2.5–4.2 V (vs. Li^+/Li) at scan rates of 0.1–0.5 mV/s. Electrochemical impedance spectroscopy (EIS) measurements are performed in the frequency range of 10 mHz–100 kHz.

3. Results and discussion

Fig. 2 illustrates the modifications of the CuO@PP separator compared to the traditional PP separator in LIBs. The CuO@PP separator exhibits increased thickness compared to the traditional PP separator. And the CuO coating layer increases both the battery capacity and high-rate discharge capability. This improvement is due to the CuO coating layer acting as an active electrode material, participating in electrochemical reactions during charge-discharge cycles by making micro contact with the cathode during cell assembly. However, ceramic material coatings such as SiO_2 and Al_2O_3 are electrochemically inert and do not contribute to battery capacity [34]. So, increasing the thickness of the CuO coating layer appropriately can further enhance the high-rate discharge capability and capacity of the battery compared to the traditional PP separator.

Fig. 3 presents the SEM images depicting the surface and cross-sectional morphologies of the two separators. Fig. 3a demonstrates that the PP separator shows a porous structure with submicron-sized pores. As seen in Fig. 3b, due to the combined effect of the PVDF binder, CuO particles are uniformly distributed on the surface of the PP separator in a regular spherical shape. This distribution creates numerous pores between the CuO particles, significantly enhancing electrolyte adsorption capacity and improving surface wettability. This indicates no aggregation of CuO and confirms the exceptional dispersal capability of the PVDF binder. Subsequently, SEM tests were conducted on the cross-sections of the PP and CuO@PP separators (Fig. 3c and d). The images indicate that the PP separator preserves its structural integrity exceptionally well. The interface between the CuO coating and the PP separator is well-preserved, demonstrating tight and consistent contact. As shown in Fig. 3d, the thickness of the CuO coating is controlled at approximately 10 μm, demonstrating precise control over the CuO coating thickness. Maintaining consistent separator thickness is crucial for the long cycle life of LIBs [35].

The wettability of the separator surface is a crucial property that determines its suitability and performance. The electrolyte was applied to two separators, and the results are presented in Fig. 4a and b. The CuO@PP separator exhibits a contact angle of 15.5°, significantly lower than the 33.55° of the PP separator, indicating excellent wetting properties. According to Fig. 4c, the porosity of the CuO@PP separator is 65.8 %, obviously higher than that of the PP separator, which is 40.1 %. The increase in the thickness of the CuO particles affects the porosity of the separator in multiple ways. Firstly, an increase in coating thickness occupies some of the pores on the separator, reducing the porosity. Secondly, increasing the thickness of the CuO coating leads to looser particle stacking, thereby increasing the porosity of the separator. Therefore, increasing the thickness of CuO particles could enhance porosity. The separator was immersed in the electrolyte for 2 h to measure its electrolyte absorption capacity. Compared to the PP separator, the CuO@PP separator showed greater electrolyte absorption. The electrolyte absorption rate for the PP separator is 76.9 %, whereas for the CuO@PP separator, it is 131.8 % (Fig. 4d). The superior liquid absorption rate of the CuO@PP separator is primarily due to the excellent hydrophilicity of CuO particles and the uniform distribution of the coating structure on the surface of the separator, which significantly enhances absorption performance. These results indicate that the CuO@PP separator demonstrates excellent electrolyte absorption and wettability, which enhances lithium-ion conductivity and facilitates their rapid conduction across the separator. The stress-strain curves indicate that the CuO@PP separator possesses a high tensile strength of approximately 100 MPa compared to the PP separator when subjected to external force (Fig. 4e). The tensile strength of the CuO@PP separator is slightly lower than that of the PP separator, which may be attributed to the increased thickness of the CuO coating, resulting in decreased tensile properties. It is also possible that the CuO coating on the separator prevents entanglement between the nanofibers.

High thermal stability ensures that separators can withstand elevated

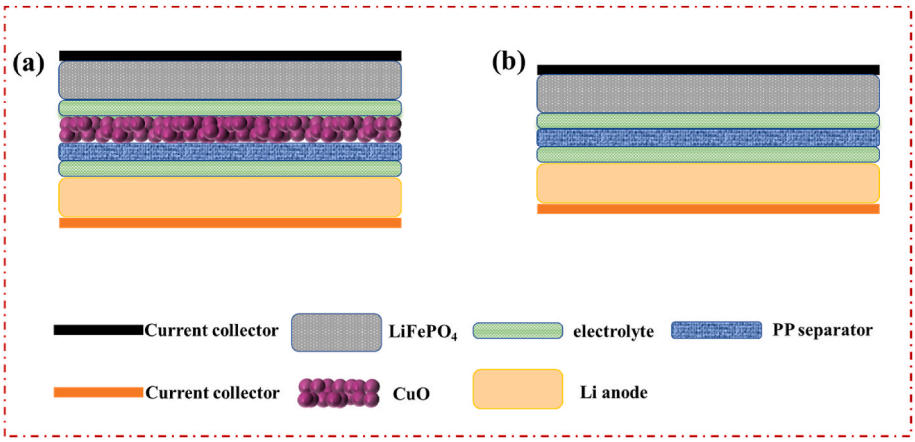


Fig. 2. Schematic illustration of (a) the CuO@PP separator; and (b) the PP separator in terms of the structure for LIBs.

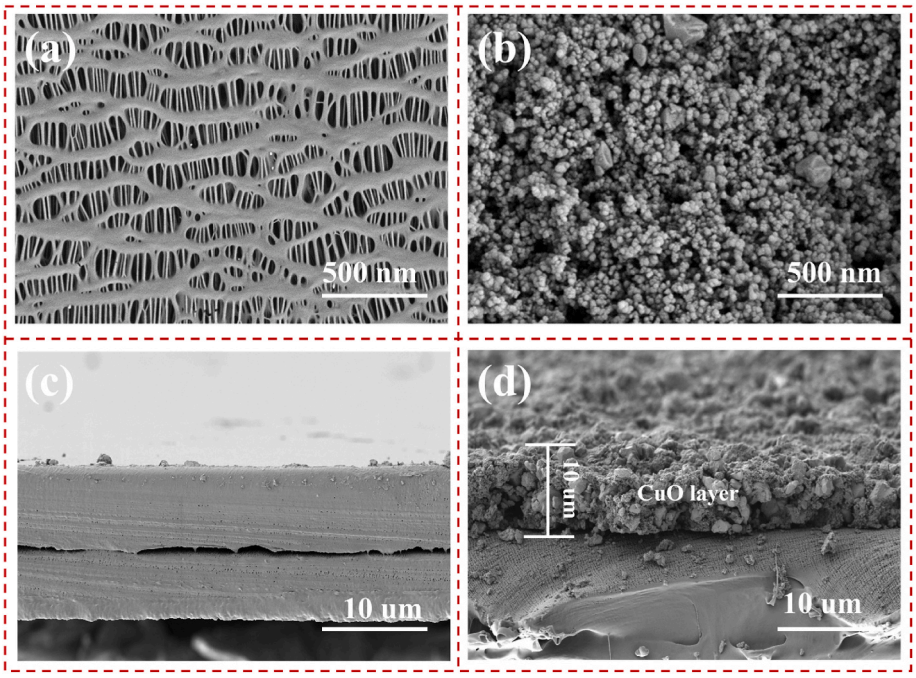


Fig. 3. (a) SEM image of the PP separator, with (c) depicting its corresponding cross-section; (b) SEM image of the CuO@PP separator, with (d) illustrating its corresponding cross-section.

temperatures without degrading or melting, which could lead to internal short circuits. Improving the thermal stability of separators is crucial to enhancing the safety performance of lithium-ion batteries. Fig. 5a illustrates the morphology of the two separators before and after 30 min of heating in an oven. The photographs in Fig. 5a depict the separators after being heated at 60, 90, 120, and 150 °C for 30 min in a vacuum oven. The area shrinkage rate of the separators is characterized by the change in area before and after heating, which is calculated by $\boldsymbol{Shrinkage}\ (\%) = \frac{S_0 - S}{S_0} \times \mathbf{100}\%$, where $\boldsymbol{S_0}$ and $\boldsymbol{S}$ are the areas of the separator before and after heating, respectively. The PP separator exhibited shrinkage with the increase of temperature. As shown in Fig. 5b, the PP separator experienced over 18 % shrinkage at 150 °C. In contrast, the CuO@PP separator demonstrated only about 5 % shrinkage under the same conditions, indicating excellent thermal stability. The thermal stability of the separator is markedly enhanced by the CuO coating, as indicated by these observations.

As illustrated in Fig. S2, we conducted comprehensive tests on batteries with varying CuO coating thicknesses to ensure accuracy. The results confirm that the CuO coating significantly enhances the fast charging and discharging capabilities of the battery. Finally, we chose a 10 μm thick CuO@PP separator as the control group for comparison with the battery assembled using the PP separator. Fig. 6a illustrates the discharge capacity of $LiFePO_4$ batteries assembled with a PP separator and a CuO@PP separator after 200 cycles at a rate of 1 C. The first discharge capacity is 145.7 mAh g^{-1} and increases to 149.2 mAh g^{-1} over 200 cycles, demonstrating an exceptional capacity retention rate of 102.4 %. In contrast, the discharge capacities of the battery prepared with the PP separator are 134.6 mAh g^{-1} in the initial cycle and 126.5 mAh g^{-1} in the 200th cycle. As shown in Fig. 6b, the evaluation of the rate performance of $LiFePO_4$/Li batteries was conducted by employing various separators at multiple current densities, ranging from 0.1 C to 10 C. Compared to the cell prepared using a PP separator, the cell prepared using a CuO@PP separator demonstrates excellent electrochemical performance at extremely high discharge rates. The battery with the CuO@PP separator maintains a relatively high discharge capacity of 122.4 mAh g^{-1}, even at an extremely high discharge rate of 10

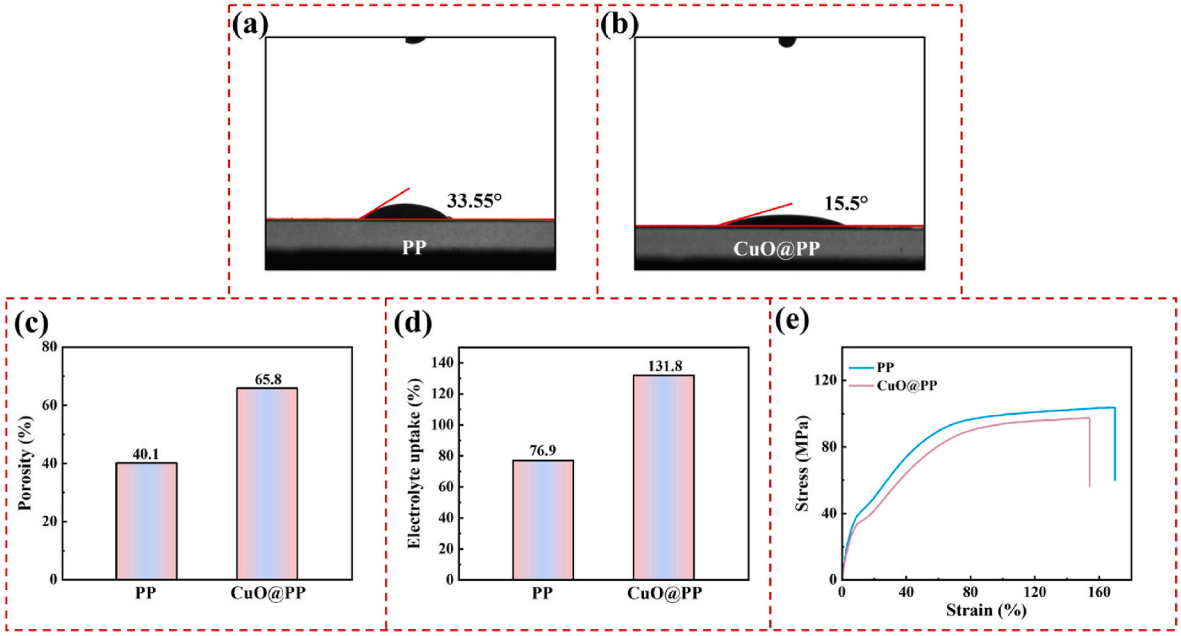


Fig. 4. Contact angle images of (a) the PP separator and (b) the CuO@PP separator after dripping the electrolyte for 10 s; (c) porosity of the PP and CuO@PP separators; (d) electrolyte uptake of the PP and CuO@PP separators; (e) stress-strain profiles of the PP and CuO@PP separators.

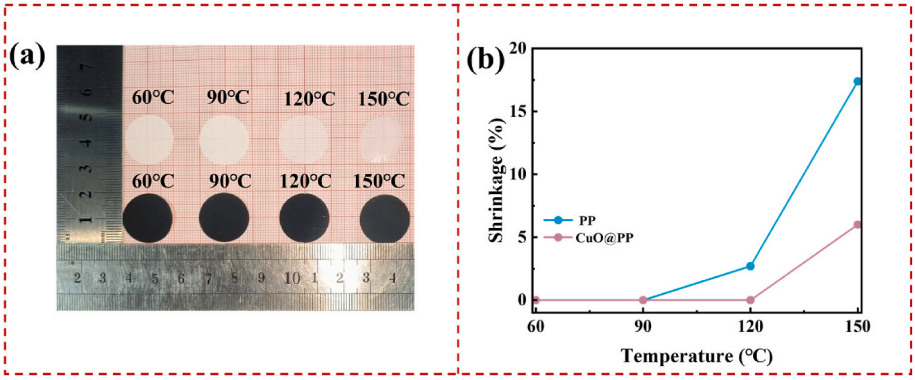


Fig. 5. (a) Photographs of the PP and CuO@PP separators after heat treatment; (b) Thermal shrinkage ratio of the PP and CuO@PP separators after being kept at 60, 90, 120 and 150 °C for 30 min.

C. Under the same conditions, the cell using the PP separator achieves only 50.2 mAh g^{-1}. This demonstrates that the CuO@PP separator enhances performance by facilitating rapid lithium-ion transport. Fig. 6c illustrates the first charge-discharge curves at 1 C, revealing that the initial discharge capacities for cells using the PP separator and the CuO@PP separator are 134.6 mAh g^{-1} and 145.7 mAh g^{-1}, respectively. Remarkably, the voltage difference between the charge and discharge platforms of batteries using the PP and CuO@PP separators represents the magnitude of the internal resistance. A smaller voltage difference indicates lower internal resistance. Therefore, the battery using the CuO@PP separator has smaller internal resistance and lower polarization than the battery using the PP separator. The EIS curves of the batteries prepared using the PP separator and the CuO@PP separator are shown in Fig. 6d. Both cells exhibit a semicircle in the mid-high frequency region, which corresponds to the charge transfer between the electrode and electrolyte interface. The semicircle diameter represents the charge transfer resistance (R_{ct}) of the battery. As shown in the figure, the R_{ct} value for the cell using a PP separator is 243.4 Ω, while the R_{ct} value for the cell using a CuO@PP separator is 114.4 Ω. Based on the differences in battery electrochemical performance shown in Fig. 6, we can conclude that the hydrophilic CuO coating significantly accelerates charge transfer and reduces interface resistance. In addition, we evaluated the electrochemical performance of the two samples at 60 °C. As illustrated in Fig. S1, the battery assembled with the CuO@PP separator maintains good cycle stability and electrochemical performance at this elevated temperature. Conversely, the capacity of the battery with the PP separator declines sharply after 25 cycles, highlighting the superior electrochemical performance of the CuO@PP separator.

Next, the properties of $LiFePO_4$ batteries with distinct separators were measured at a current density of 10 C. As shown in Fig. 6e, over 400 cycles, the capacity retention rate of the $LiFePO_4$ cell prepared using a PP separator remains relatively stable. The first discharge capacity is only 43.8 mAh g^{-1}, increasing to 46.1 mAh g^{-1} after 400 cycles. This indicates that the PP separator is unsuitable for high-rate charge and discharge applications. The cell with the CuO@PP separator demonstrates outstanding electrochemical performance during 10 C high-rate discharge. The first discharge capacity reaches 129.8 mAh g^{-1}, and over 400 cycles, it retains 120.0 mAh g^{-1}, representing a capacity retention rate of 92.4 %. Additionally, the battery maintains exceptional high-rate cycling stability, exhibiting minimal degradation after 400 charge and discharge cycles. Batteries assembled with CuO@PP separators exhibit better high-rate discharge capability compared to those

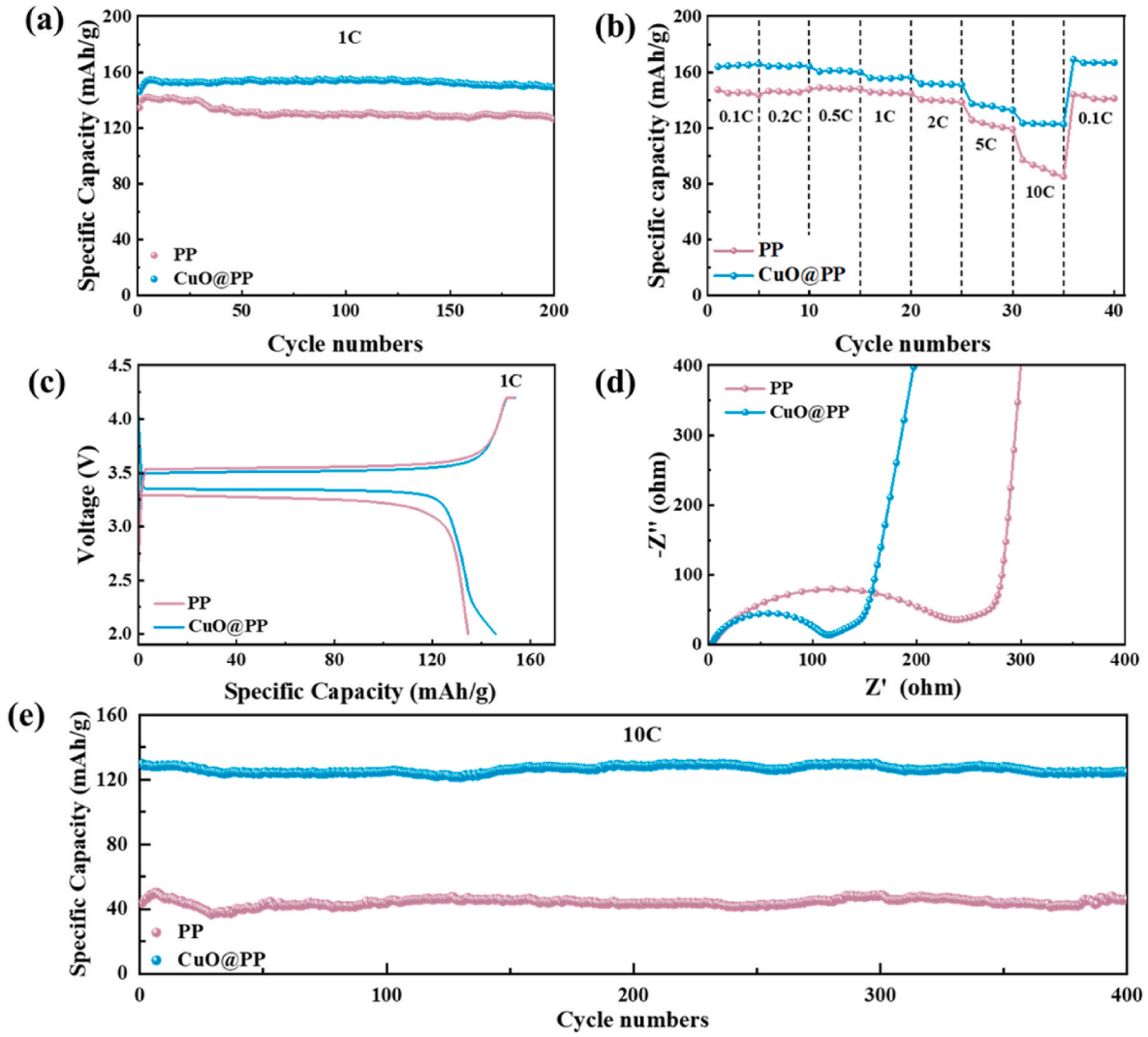


Fig. 6. Electrochemical performances of LFP/Li half cells with PP and CuO@PP separators: (a) 1 C cycling capability; (b) rate performances at various discharge rates between 0.1 and 10 C; (c) the first charge-discharge curves at 1 C; (d) EIS plots. (e) cycling stabilities combined of PP and CuO@PP separators at 10 C.

with PP separators. There are two possible reasons for the improvement in rate performance. First, the CuO coating on the separator improves the wettability of the interface, allowing more electrolytes to penetrate and fully immerse the diaphragm, thereby enhancing the diffusion rate of lithium ions. Second, the CuO coating reacts with the cathode electrode to form a cathode electrolyte interphase (CEI) similar to that of the anode electrode, enhancing interface conductivity and providing additional capacity. Compared to the PP separator, the CuO@PP separator performs better in high-rate discharge capability, offering a new approach for the quick charging and discharging of LIBs.

The CV curves of the assembled battery were evaluated at scan rates ranging from 0.1 to 0.5 mV/s, and the diffusion coefficient of lithium ions was subsequently calculated. As shown in Fig. 7 (a, b, d, and e), as the scan rate increases, the peak current (I_p) demonstrates a rising trend. Additionally, the oxidation peak shifts to a higher potential, while the reduction peak shifts to a lower potential. As expected, the CuO@PP separator demonstrates the smallest potential difference between the two peaks, suggesting reduced polarization and enhanced electrochemical reversibility. Furthermore, the lithium-ion diffusion coefficient (D_{Li+}) can be determined using the Randles-Sevcik equation [36, 37]:

$$I_p = 2.687 \times 10^5 n^3 D^{1/2} A C \gamma^{1/2} \quad (3)$$

where n represents the number of electrons transferred in the Fe^{2+}/Fe^{3+} redox couples, D represents the Li^+ diffusion coefficient, A represents the electrode surface area, C represents the lithium-ion concentration and γ represents the scan rate. Fig. 7b and e illustrate the relationship between the current peak (I_p) and scan rate ($\gamma^{1/2}$). The D_{Li+} of the CuO@PP separator is 1.04×10^{-11} cm^2 s^{-1} at the reduction peak and 8.76×10^{-12} cm^2 s^{-1} at the oxidation peak. Both values are higher than the D_{Li+} of the PP separator, which are 9.06×10^{-12} cm^2 s^{-1} at the reduction peak and 5.71×10^{-12} cm^2 s^{-1} at the oxidation peak, demonstrating that the CuO@PP separator improves the diffusion kinetics, thereby enhancing the electrochemical performance.

Fig. 7a illustrates that the battery equipped with a PP separator experiences a marked current increase at 3.5 V on the CV curve, indicating the oxidation of $LiFePO_4$ and the formation of an oxidation peak. Conversely, the battery with a CuO@PP separator exhibits a significant current not only at 3.5 V but also at 3.0–3.3 V (as shown by the dashed box). Subsequently, a sloping line in the charge-discharge curve, corresponding to the current change at 3.0–3.3 V in the CV curve, becomes the charging platform period of $LiFePO_4$ (Fig. 7d). From the analysis of the discharge curve, it is evident that as the discharge process nears completion, the curve shifts to the right, resulting in an increased discharge capacity. Therefore, the CuO coating on the separator can be inferred to engage in the electrochemical reaction within the voltage

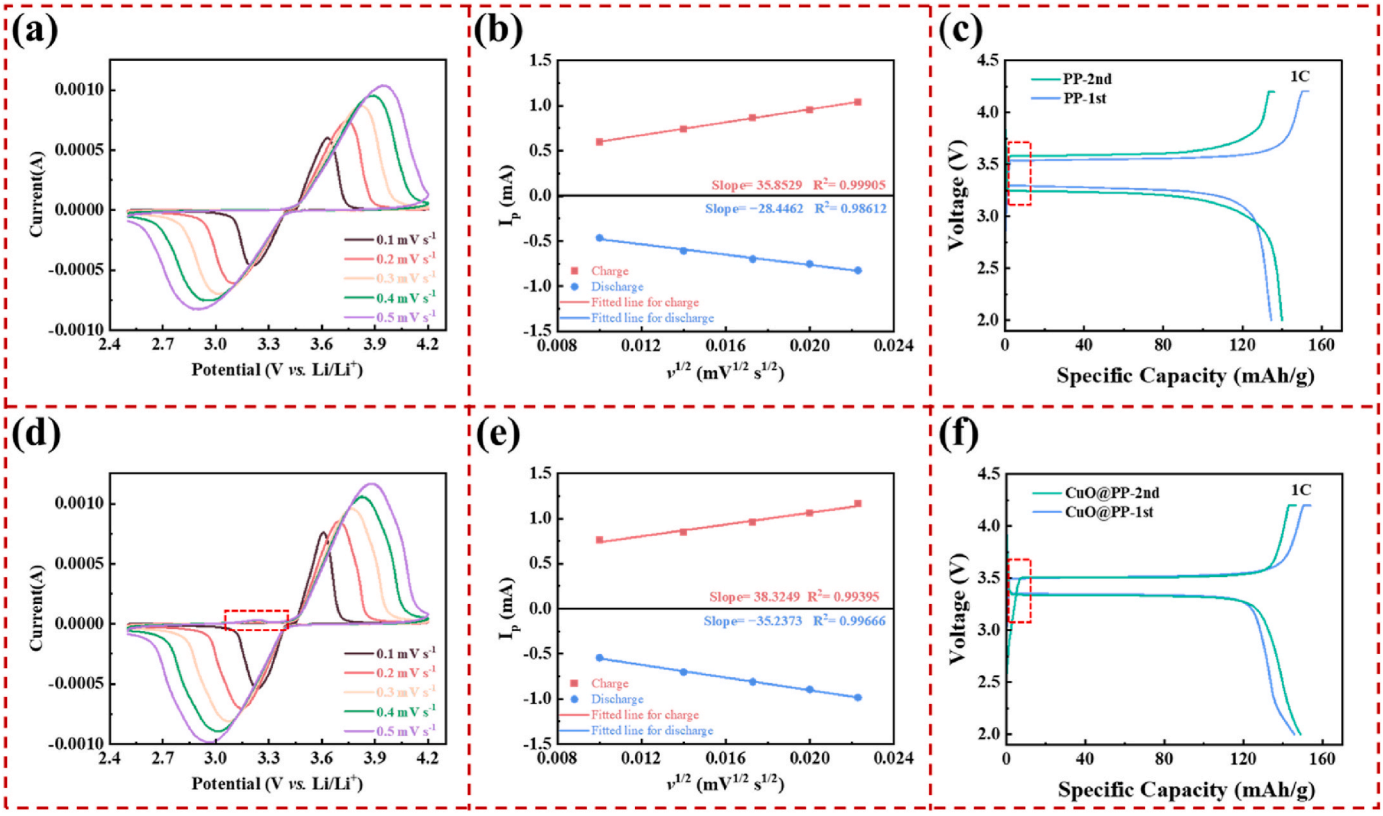


Fig. 7. (a, d) CV profiles at scan rates of 0.1, 0.2, 0.3, 0.4, and 0.5 mV/s of PP, and CuO@PP separators; (b, e) Liner relation of the cathode peak current (I_p) and the square root of the scan rate ($\gamma^{1/2}$) for intercalation and deintercalation process; (c) The first and second charge and discharge curves of PP separator; (f) The first and second charge and discharge curves of CuO@PP separator.

range of 3.0–3.3 V. It is noteworthy that the width of the discharge platform (approximately 130 mAh g^{-1}) has remained largely unchanged. This indicates that the discharge capacity contributed by the cathode $LiFePO_4$ is similar for cells prepared using CuO@PP and PP separators. The increased discharge capacity observed in the battery assembled with CuO@PP (around 10 mAh g^{-1}) may be attributed to the contribution of the CuO coating participating in the reaction. Lithium ions may have electrochemically reacted with the CuO coating [27,38,39], as illustrated in the following chemical equation:

$$CuO + xLi^+ + xe \leftrightarrow Li_xCuO \quad (4)$$

In the initial cycle, the charge-discharge curve of the CuO@PP separator closely resembles that of the cell with the PP separator (Fig. 7c and f). However, when discharging in the second cycle, as shown in the dashed box in Fig. 7f, the charge capacity of the CuO@PP separator at 3.0–3.5 V is 10 mAh g^{-1} higher than that of the PP separator. This may indicate that some Li^+ ions react in this range to generate Li_xCuO. The incorporation of Li_xCuO can improve the conductivity and capacity at the interface, resulting in superior electrochemical performance of the battery [38,40–42].

In summary, coatings typically increase separator thickness, which detrimentally affects the electrochemical properties of LIB when the combined thickness of the cathode electrode, anode electrode, and separator is considered. Although certain studies suggest that specific coatings can marginally enhance battery capacity, these improvements are insufficient to counteract the adverse effects of increased separator thickness, ultimately diminishing electrochemical performance. However, the CuO coating significantly enhances discharge capacity despite the augmented separator thickness. Additionally, the CuO coating enhances electrolyte penetration and actively participates in reactions during the charge and discharge processes. This contribution ensures excellent electrochemical performance in the battery, making it superior to other coatings [32,43,44].

4. Conclusion

By modifying the surface of the PP separator with CuO, a functional active electrode material, we have developed a separator that enhances electrochemical performance in lithium-ion batteries and investigated its potential application. Enhanced interfacial properties between the electrolyte and separator were observed, indicating that the addition of CuO particles improves electrolyte absorption. EIS results showed that the CuO@PP separator exhibits lower impedance, thereby contributing to improved electrochemical performance. In $LiFePO_4$/Li batteries, the CuO@PP separator demonstrated exceptional cyclic stability. Remarkably, the discharge capacity increased after 200 charge-discharge cycles, further highlighting its superior performance. Furthermore, the CuO@PP separator demonstrated a superior lithium ions transport number and electrochemical stability. Notably, it exhibited exceptional performance at high rates, sustaining a capacity of 120.0 mAh g^{-1} over 400 cycles, with the cells maintaining a high-capacity retention of 92.4 % at a rate of 10 C. Thus, enhancing the reliability and performance of LIBs by modifying the PP separator with a CuO coating presents a promising approach and offers novel perspectives for designing and manufacturing advanced battery separators.

CRediT authorship contribution statement

Yajie Yuan: Writing – original draft, Data curation. **Yin Li:** Writing – review & editing, Validation, Methodology, Investigation, Funding acquisition, Conceptualization. **Junxian Hu:** Supervision, Methodology. **Keyu Zhang:** Supervision. **Shaoze Zhang:** Supervision. **Bin Yang:** Supervision. **Yaochun Yao:** Supervision, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jpowsour.2024.235412.

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